

## SUMMARY

Psychology PhD candidate specializing in quantitative methods: nonlinear time-series analysis, Monte Carlo simulation, and formal computational modeling of intensive longitudinal behavioral data. Builds reproducible, parallelized analysis pipelines in R and has full-stack app development (JavaScript, Node.js, React Native) and applied user research experience from industry.

## WORK EXPERIENCE

**Research Associate**, *University of Hamburg* Feb 2024 – present

- Analyzing intensive longitudinal data from 2,000+ participants across 18 cross-national datasets with nonlinear time-series methods in R, including a reproducible harmonization pipeline and parallelized computing
- Developing and validating formal computational models of emotion dynamics using Monte Carlo simulation
- Designing, conducting, and analyzing behavioral experiments, including simulation-based power analyses
- Building version-controlled (Git), reproducible analysis workflows for collaborative research
- Designing and teaching the seminar "Personality as a Dynamic Process"
- Supervising a research assistant, including task delegation, code review, and quality control of data collection

**Frontend App Developer (part-time, 8h/week)**, *HMU Health and Medical University Potsdam* Oct 2023 – present

- Developing the front end and user flow of the personality-psychology guessing game app *Who Knows* (12,000+ users)

**Fullstack App Developer (part-time, 8h/week)**, *University of Münster* Sep 2020 – Jan 2023

- Developed the server side of *Who Knows* with Node.js/Sequelize: set up progressive web app (PWA) architecture, API route handling, user authorization, and database management
- Built the counseling app *Kleine Kinder – Große Gefühle* full-stack: implemented the design in the front end, API route handling, and database management

**User Researcher (Intern)**, *Aumio, Berlin* Jan 2021 – Mar 2021

- Developed a remote usability test and beta testing program
- Conducted surveys and interviews and aggregated results in affinity maps
- Conducted basic accessibility testing

**User Experience Coach (Intern)**, *artop, Berlin* Nov 2020

- Designed seminars on user experience
- Assisted in giving seminars

## SKILLS

**Programming & tools:** R, Python, SQL, JavaScript, React Native, Redux Toolkit, Vue.js, Git

**Statistics & methods:** Time-series analysis, Monte Carlo simulation, multilevel modeling, empirical dynamic modeling, Bayesian modeling (Stan, JAGS), network analysis (qgraph, igraph), parallel computing (furrr), research design, data visualization (ggplot2)

**Languages:** German (native), English (fluent), Spanish (conversational), Italian (beginner)

## EDUCATION

**PhD Psychology**, *University of Hamburg* Feb 2024 – present

- Dissertation: *Emotions as Complex Systems: Implications for Theory, Modeling, and Experience Sampling Research*

**Research Exchange**, *University of Melbourne* Jan 2026 – Apr 2026

- Funded by a DAAD research grant for doctoral candidates

**M.Sc. Psychology – Cognitive Neuroscience**, *University of Münster* Oct 2020 – Jan 2023

- Final grade: 1.2
- Thesis: *Estimating Crossed Random Effects in Binomial GLMMs With Planned Missingness: How Many Players Need to Play the Guessing Game App Who Knows?* (1.0)

**Erasmus Exchange**, *Universidad Complutense de Madrid* Sep 2020 – Jan 2021

**B.Sc. Psychology**, *University of Münster* Oct 2017 – Sep 2020

- Final grade: 1.5
- Thesis: *"Just a good feeling"? – On the relationship between watching football and mental well-being* (1.6)

**B.Sc. Material Science**, *RWTH Aachen* Oct 2015 – Sep 2017

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## SELECTED PUBLICATIONS

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- Kueppers, S.**, Zickfeld, J. H., Ghadertootoonchi, S., & Lange, J. (2026). Explaining emotion co-occurrence: Comparison of parsimonious, formal theories using multimodal, within-person data. *Emotion*. Advance online publication. <https://doi.org/10.1037/emo0001740>
- Rau, R., **Kueppers, S.**, Back, M. D., & Grosz, M. P. (2026). Item features affect how well people can predict self-reported personality traces of strangers. *European Journal of Personality*, 1-20. <https://doi.org/10.1177/08902070261467481>
- Kueppers, S.**, Rau, R., & Scharf, F. (2024). Using Monte Carlo simulation to forecast the scientific utility of psychological app studies: A tutorial. *Multivariate Behavioral Research*, 59(4), 879-893. <https://doi.org/10.1080/00273171.2024.2335411>